Long-range epigenetic concordance

a per-cell, per-bin signal that separates cell states even when mean methylation / accessibility is identical

Single-cell NOMe-HiC contact

chromatin contact

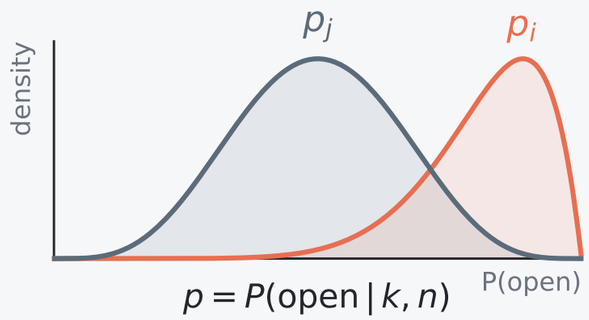


locus *i* — k_i/n_i

locus *j* — k_j/n_j

GCH accessibility · HCG (CpG) methylation

Per-end state posterior



$$p = P(\text{open} \mid k, n)$$

Beta–Binomial mixture (EM)

Soft pairwise co-state

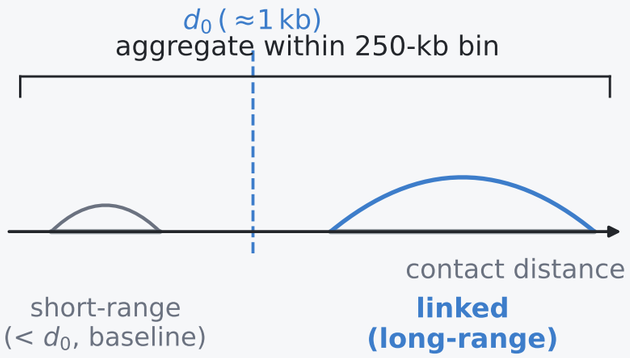
co-open
$p_i\, p_j$

co-closed
$(1-p_i)(1-p_j)$

mixed
$p_i(1-p_j) + (1-p_i)p_j$

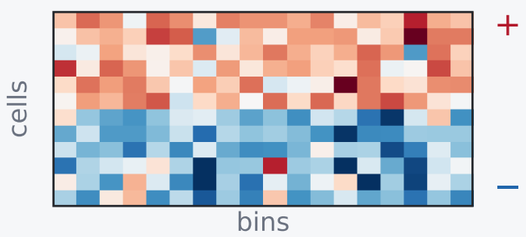
soft expected counts (no thresholding)

Long-range vs local baseline



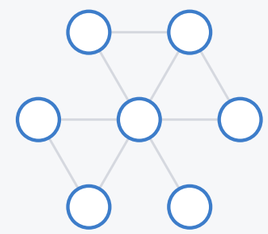
Signed long-range concordance

$$\Delta_{\text{signed}} = (\hat{\pi}_{\text{co-open}} - \hat{\pi}_{\text{co-closed}})_{\text{linked}} - (\hat{\pi}_{\text{co-open}} - \hat{\pi}_{\text{co-closed}})_{\text{short-range}}$$



Cell-graph denoising (RWR)

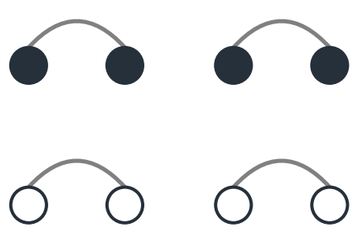
random-walk-with-restart smoothing
over the cell *k*-NN graph



quantile-normalised, imputed → UMAP

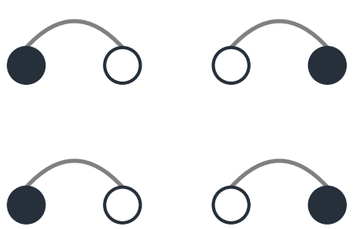
Same mean methylation / accessibility, opposite long-range concordance

cell state 1



concordance = 1.0

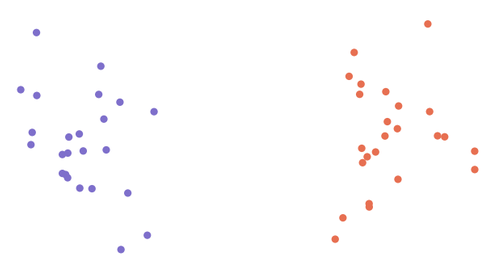
cell state 2



concordance = 0.0

workflow →

UMAP: signed concordance



the two states separate

● methylated / accessible ○ unmethylated / closed

each loop = two loci in contact within one cell